Technical Documentation

Introduction

RAG combines retrieval systems with language models to provide accurate and context-based answers.

Use Case

Customer support automation for return policy, tracking, delivery, and support queries.

Working Process

- Load PDF
- Convert into chunks
- Create embeddings
- Store in vector DB
- Accept user question
- Retrieve relevant context
- Generate final answer

LangGraph Usage

Controls workflow execution between nodes and manages state transitions.

HITL Benefits

Allows human review for complex or uncertain queries.

Challenges & Trade-offs

Speed vs accuracy, chunk size vs context quality, API cost vs performance.

Testing Queries

- What is return policy?
- How can I track my order?
- How to contact support?

Future Enhancements

Multi-document support, multilingual chatbot, memory-based chat, cloud deployment.